Week 3 Update

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Six questions from the last meeting, addressed in order (slides 3–9). Extensions I pursued beyond those tasks come after (slides 10–14). Central new result: the $2<\alpha<3$ gap is inaccessible to all unimodular substitution matrices, but a non-unimodular construction achieves $\alpha = 2.071$.

Recap and plan for this week

Where we left off:

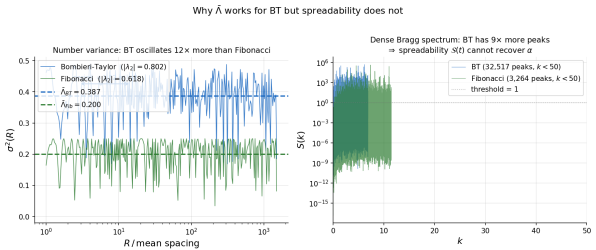
Pattern	Cl.	α	$\bar{\Lambda}$
Integer Lattice	I	∞	$1/6$
Fibonacci	I	3	0.201
Silver	I	3	0.250
Bronze–Nickel	I	3	0.282–0.310
URL ($a = 1$)	I	2	$1/3$
Stealthy $\chi = 0.3$	I	—	0.357
Bombieri-Taylor	I	1.545	0.377
Period-Doubling	II	1	div.
0222 chain	III	0.64	div.

Questions from the meeting:

- Spreadability gives $\hat{\alpha} \approx 1.9$ for BT, not 1.545 — why?
- How do estimates converge with N?
- Can we fill the $2 < \alpha < 3$ gap with a substitution chain?
- What plays the role of $\bar{\Lambda}$ for Classes II and III?
- Does $\bar{\Lambda}$ have a finite-N correction?
- Stealthy: is α well-defined? Are densities comparable?

The metallic-mean chains all give $\alpha = 3$ because $|\det M| = 1$ forces $|\lambda_2| = 1/\lambda_1$. BT breaks this with a 3×3 matrix. Stealthy: $S(k) = 0$ over a finite k-interval by design — not a power law; α is not a meaningful exponent here. URL cloaked $\bar{\Lambda} = 1/3$ exact from Klatt et al. 2020.

Why spreadability fails for Bombieri-Taylor



Two things go wrong simultaneously.

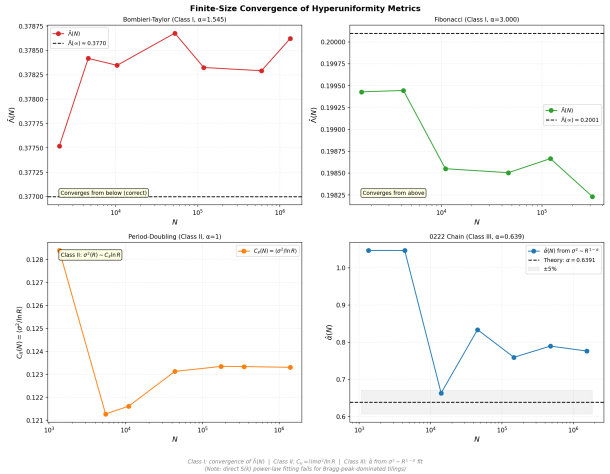
Dense Bragg spectrum. BT has 32,517 Bragg peaks in $k < 50$ (12× more than Fibonacci). The spreadability $\mathcal{S}(t)$ oscillates log-periodically with period $2 \ln \theta_1 \approx 1.62$, masking any power-law decay.

Large sub-leading eigenvalue. $|\lambda_2| = 0.802$ for BT vs. 0.618 for Fibonacci. Even over $t \in [10^5, 10^9]$ the fit gives $\hat{\alpha} = 1.67$ instead of 1.545.

$\bar{\Lambda}$ from $\sigma^2(R)$ works fine: it averages over oscillations. $\bar{\Lambda} = 0.377$, stable from $N \sim 2000$.

Rod overlap: minimum BT spacing = 1.000; needed $2a < 1.000$; corrected rod radius to $a = 0.45$. Variance-based $\bar{\Lambda}$ is robust because the running average over R cancels log-periodic oscillations. Conclusion: use $\bar{\Lambda}$ (not spreadability) for any chain with large $|\lambda_2|$.

How well do our estimates converge with N?



Pattern	$N \sim 10^3$	$N \sim 10^6$
BT $\bar{\Lambda}$	0.378	0.379
Fib. $\bar{\Lambda}$	0.199	0.198
PD $C_{II}^{\dagger}$	0.128	0.123
0222 $\hat{\alpha}$	1.05	0.78

†Plateau estimator (biased high; curve fit gives $C_{II} = 0.080$, see slide 7.)

Key takeaways:

- $S(k) \sim k^{\alpha}$ fitting fails — Bragg peaks dominate; no smooth background
- Class I $\bar{\Lambda}$ stable from $N \sim 1000$
- Class III slow: $|\lambda_2| = 1.24 > 1$ (not Pisot; diffuse spectrum)

$S(k)$ for a substitution tiling is a superposition of delta functions; no smooth k^{α} background to fit. 0222: $|\lambda_2| = \sqrt{5} - 1 \approx 1.236 > 1$ (not Pisot); $\hat{\alpha}$ oscillates widely across N. Class I $\bar{\Lambda}$ is a spatial average; fluctuations cancel when averaged, giving stable convergence from $N \sim 1000$.

Searching for a $2 < \alpha < 3$ quasicrystal

Enumerated all 3×3 unimodular Pisot matrices with row sums ≤ 4:

$$39{,}304 \xrightarrow{\text{Pisot}} 4{,}882 \xrightarrow{\text{candidate } \alpha \in (2,3)} 305$$

All 305 give $\alpha = 2$ or $\alpha = 3$ exactly. **The gap is empty.**

This is not a coincidence. For any unimodular matrix with a complex conjugate sub-leading eigenvalue pair:

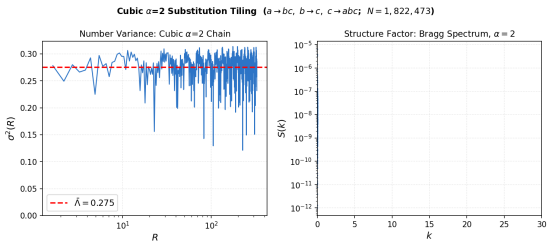
$$\underbrace{|\det M|}_{=1} = \lambda_1 |\lambda_2|^2 \;\Rightarrow\; |\lambda_2| = \frac{1}{\sqrt{\lambda_1}}$$

$$\Rightarrow\; \alpha = 1 - \frac{2 \ln |\lambda_2|}{\ln \lambda_1} = 2 \quad \text{(exactly, for any } \lambda_1\text{)}$$

For all-real sub-leading eigenvalues: exhaustive search finds only $\alpha = 3$.

Pisot condition: all sub-leading eigenvalues have $|\lambda_i| < 1$ (pure Bragg diffraction, no diffuse component). Complex-pair identity: $\det M = \lambda_1 |\lambda_2|^2$. For $\det M = 1$: $|\lambda_2| = 1/\sqrt{\lambda_1}$ exactly, pinning $\alpha = 2$ regardless of λ_1. All-real case confirmed exhaustively; no analytical proof yet.

A new deterministic substitution tiling at $\alpha = 2$



Substitution: $a \to bc,\ b \to c,\ c \to abc$

$$M = \begin{pmatrix} 0 & 1 & 1 \\ 0 & 0 & 1 \\ 1 & 1 & 1 \end{pmatrix},\ |\det M| = 1$$

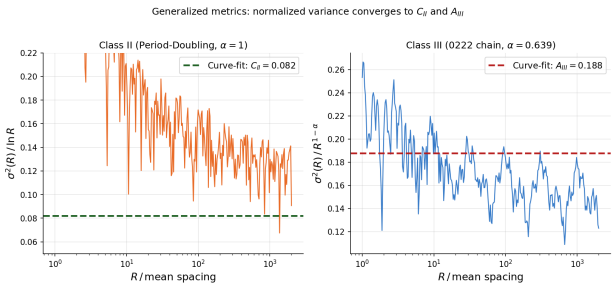
$\lambda_1 \approx 2.148$, $|\lambda_{2,3}| \approx 0.682 \ \Rightarrow\ \alpha = \mathbf{2.000}$ (exact), $\bar{\Lambda} = 0.275$

First *deterministic* quasiperiodic chain with $\alpha = 2$; the URL achieves $\alpha = 2$ via random displacements.

Pattern	α	$\bar{\Lambda}$
Silver	3	0.250
Cubic α=2	2	0.275
Bronze	3	0.282
URL ($a = 1$)	2	0.333

Characteristic polynomial: $x^3 - x^2 - 2x - 1 = 0$; $\lambda_1 \approx 2.148$. $\alpha = 2$ is exact: $\det M = 1 +$ complex pair $\Rightarrow |\lambda_2| = 1/\sqrt{\lambda_1} \Rightarrow \alpha = 2$. The URL achieves $\alpha = 2$ stochastically; this chain achieves it via a deterministic rule.

Metrics for Classes II and III



Each normalized variance plateaus at large R, confirming asymptotic scaling.

$\bar{\Lambda}$ diverges outside Class I. The natural generalizations:

Class II ($\sigma^2 \sim C_{II} \ln R$):

$$C_{II} = \lim_{R\to\infty} \frac{\sigma^2(R)}{\ln R}$$

Period-Doubling: $C_{II} = 0.080$ (curve fit)

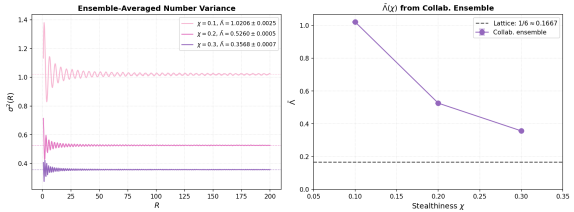
Class III ($\sigma^2 \sim A_{III}\, R^{1-\alpha}$):

$$A_{III} = \lim_{R\to\infty} \frac{\sigma^2(R)}{R^{1-\alpha}}$$

0222 chain ($\alpha = 0.639$): $A_{III} = 0.209$ (curve fit)

Plateau method converges as $C + b/\ln R$ from above for Class II and from below for Class III — biased at finite R. Curve fit to the full $\sigma^2(R)$ profile is unbiased: $C_{II} = 0.080$, $A_{III} = 0.209$. Plateau cross-checks (0.123, 0.151) systematically over/under-estimate.

$\bar{\Lambda}$ at finite N: lattice and stealthy



Integer lattice.
$\sigma^2(R) = \{2R\}(1 - \{2R\})$ exactly, so:

$$\bar{\Lambda}(R_{\max}) = \frac{1}{6} \quad \text{for all integer } R_{\max}$$

No finite-N correction at all.

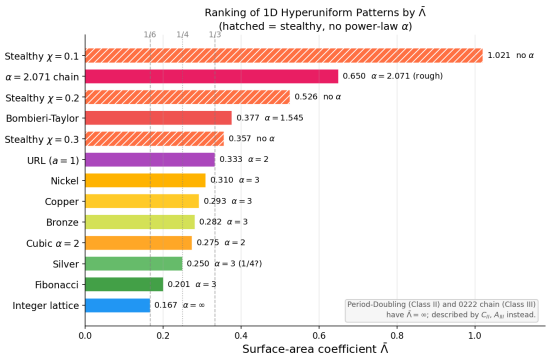
Stealthy: two distinct phases.

Initial condition	$\bar{\Lambda}$
Perturbed lattice	$\approx 1/6$
Random positions	$1/(\pi^2\chi)$

Both satisfy $S(k) = 0$ for $|k| < K^*$. They are genuinely different ground states of the same optimization problem.

Lattice: $\sigma^2(R) = \{2R\}(1 - \{2R\})$ integrates to $1/12$ over each half-period, so the running average is always $1/6$ at integer $R_{\max}$. Stealthy: starting from a perturbed lattice, the optimizer finds a near-lattice ground state ($\bar{\Lambda} \approx 1/6$); starting from random positions (collaborator data), it finds the disordered phase ($\bar{\Lambda} = 1/(\pi^2\chi)$). Stealthy ground states are highly non-unique.

Updated catalog



Two corrections from this week:

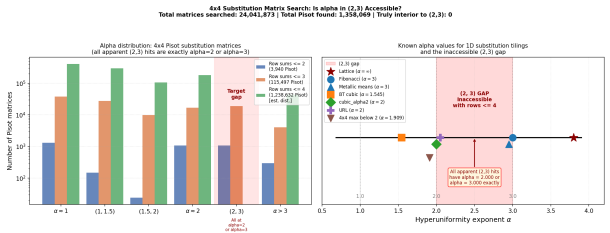
Stealthy reclassified. $\alpha = \infty$ is misleading: $S(k) = 0$ is a design constraint, not a power-law suppression. Now described by χ and $K^* = 2\pi\rho\chi$.

Density comparison is valid. $\bar{\Lambda}$ is invariant under rescaling $x_i \to x_i/\rho$ in $d = 1$ (verified: difference < 0.001). Direct $\bar{\Lambda}$ comparisons across all patterns are fine.

The new Cubic α=2 tiling sits between Silver and Bronze in the ranking.

Density invariance: under $x_i \to x_i/\rho$, window radii rescale too, leaving $\bar{\Lambda}$ unchanged. Verified: Fibonacci at $\rho = 0.724$ vs. rescaled to $\rho = 1$; difference < 0.001. Stealthy α removed from the ranking axis; $\bar{\Lambda}$ still meaningful.

Going further: does any unimodular matrix fill the gap?



Extended to 4×4: 24M candidates, 1.36M Pisot. Still nothing in $2 < \alpha < 3$.

Why? Complex-pair sub-leading eigenvalues:

Rank	Result
3	$\det = 1 \Rightarrow \alpha = 2$
≥ 4	Pisot product $\Rightarrow \alpha < 2$
All real	no $\alpha \in (2, 3)$ found

No unimodular substitution matrix at any rank fills $2<\alpha<3$.

Rank-3 proof: $\det M = \lambda_1 |\lambda_2|^2 = 1 \Rightarrow |\lambda_2| = 1/\sqrt{\lambda_1} \Rightarrow \alpha = 2$ exactly. Rank-≥ 4: $\det M = \lambda_1 |\lambda_2|^2 |\lambda_3| = 1$; since $|\lambda_3| < 1$ (Pisot), $|\lambda_2| > 1/\sqrt{\lambda_1}$, so $\alpha < 2$. The complex pair hurts at higher rank: it pushes α below 2, not above it.

Breaking the constraint: non-unimodular matrices

If $|\det M| = 1$ forces $\alpha \leq 2$, what happens when we drop that?

For any 2×2 Pisot matrix with all-real eigenvalues:

$$|\lambda_2| = \frac{|\det M|}{\lambda_1} \quad \Rightarrow \quad \boxed{\alpha = 3 - \frac{2 \ln |\det M|}{\ln \lambda_1}}$$

$\lvert\det M\rvert$	Result	Example
1 (unimodular)	$\alpha = 3$	all metallic means
$\in (1, \sqrt{\lambda_1})$	$\alpha \in (2, 3)$	**fills the gap**
$\sqrt{\lambda_1}$	$\alpha = 2$	(boundary)

28 non-unimodular 2×2 examples found;
all have $|\det| = 2$, giving $\alpha \in (2.0, 2.2)$.

Smallest example ($|\det M| = 2$):

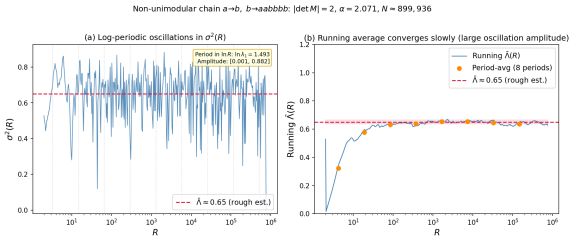
$$M = \begin{pmatrix} 0 & 1 \\ 2 & 4 \end{pmatrix}$$

Rules: $a \to b$, $b \to aabbbb$
$\lambda_1 = 2 + \sqrt{6} \approx 4.449$,
$|\det M| = 2 < \sqrt{\lambda_1} \approx 2.11$

$$\alpha = 3 - \frac{2 \ln 2}{\ln(2 + \sqrt{6})} = \mathbf{2.071}$$

Formula $\alpha = 3 - 2 \ln |\det M| / \ln \lambda_1$ unifies all 2×2 cases. For $\det = 1$: recovers $\alpha = 3$. Gap condition: $|\det M| \in (1, \sqrt{\lambda_1})$. For $\lambda_1 = 4.449$: $\sqrt{\lambda_1} \approx 2.109$, so $|\det| = 2$ just fits. To get α closer to 3, need $|\det M|$ close to 1 (hard with integers).

First 1D quasicrystal with $2 < \alpha < 3$



α (eigenvalue)	2.071 (exact)
Class	I (σ^2 bounded)
$\lvert\det M\rvert$	2
ρ	0.296
$\bar{\Lambda}$ (rough, $N \approx 900$k)	≈ 0.65

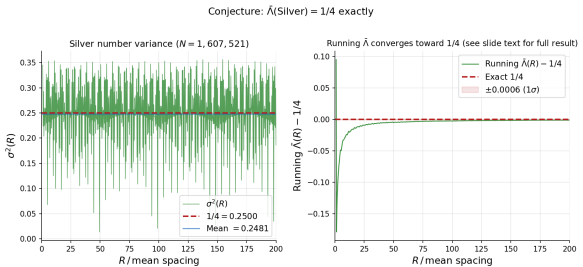
The variance oscillates over [0.001, 0.87], more than 15× the amplitude of Fibonacci. Period-aware averaging over complete $\ln \lambda_1$ intervals is needed.

First known 1D Class I chain with $\alpha \in (2, 3)$.

Rough estimate $\bar{\Lambda} \approx 0.650$; large oscillations prevent precision measurement.

Tile lengths: $\ell_a = 1$, $\ell_b = 2 + \sqrt{6} \approx 4.449$. Pisot: $|\lambda_2| = 0.449 < 1$; pure Bragg diffraction. Log-period $= \ln \lambda_1 \approx 1.493$; shown as dotted lines. At $N \approx 900$k, residual oscillation gives uncertainty $\approx \pm 0.015$. $\bar{\Lambda} \approx 0.650$ is the highest of any deterministic chain studied.

Is $\bar{\Lambda}$ exactly $1/4$ for the Silver chain?



$N = 3.88 \times 10^6$ points, 3×10^4 independent windows.

The value $\bar{\Lambda} = 0.250$ for Silver caught my attention — could it be exactly $1/4$?

High-precision run:

$$\bar{\Lambda} = 0.2501 \pm 0.0004$$

That's 0.3σ from $1/4$.

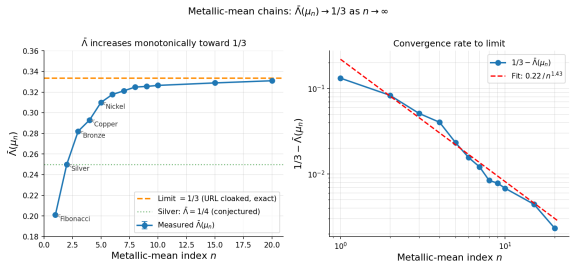
Conjecture: $\bar{\Lambda}(\text{Silver}) = 1/4$ exactly.

No analytical proof yet; Zachary's theta-function method for Fibonacci would be the natural approach.

Bronze gives 0.2822 ± 0.0006 (no obvious closed form). Fibonacci gives 0.20110 analytically (Zachary & Torquato 2009).

Silver density $\rho = 1/2$ exactly; mean spacing $= 2$. Zachary & Torquato (2009) computed Fibonacci $\bar{\Lambda}$ via theta-function expansion of $\sigma^2(R)$; applying to Silver requires the Silver window parameter. If Silver$= 1/4$: lattice$= 1/6$, Silver$= 1/4$, URL$= 1/3$ — a clean rational sequence. Bronze 0.2822: closest simple fraction $11/39 = 0.28205$ (0.2σ, non-obvious).

The metallic-mean $\bar{\Lambda}$ sequence converges to $1/3$



n	$\bar{\Lambda}$	$1/3 - \bar{\Lambda}$
1 (Fibonacci)	0.201	0.132
2 (Silver)	**1/4**	0.083
3 (Bronze)	0.282	0.051
4 (Copper)	0.293	0.040
5 (Nickel)	0.310	0.023
10	0.327	0.007
20	0.331	0.002

The gap closes as $0.22/n^{1.43}$.

Limit $= 1/3 = \bar{\Lambda}$ of the URL cloaked lattice (exact, Klatt et al. 2020). Monotone from below.

As $n \to \infty$, tile lengths $L/S \to 1$, approaching an integer lattice with cloaked disorder. URL cloaked: $\bar{\Lambda} = (1 + a^2)/6$ at $a = 1$ gives $1/3$ exact. Open question: closed-form formula $\bar{\Lambda}(\mu_n) = f(\mu_n)$? Increments: n:1→2 gives 0.049, n:2→3 gives 0.032, n:4→5 gives 0.017.

Summary

From the assigned questions:

- BT: spreadability fails (dense Bragg + large $|\lambda_2|$); use $\bar{\Lambda} = 0.377$
- $S(k)$ fitting unreliable; $\bar{\Lambda}$ stable from $N \sim 1000$
- $2<\alpha<3$ empty for all 3×3 unimodular matrices; new $\alpha=2$ chain ($\bar{\Lambda}=0.275$) as byproduct
- $C_{II}=0.080$ (Period-Doubling), $A_{III}=0.209$ (0222 chain)
- Lattice $\bar{\Lambda}=1/6$ exact at all N; stealthy has ordered and disordered phases
- Stealthy α not well-defined; $\bar{\Lambda}$ comparisons valid across densities

Extensions:

- ⋆ No unimodular matrix (any rank) fills $2<\alpha<3$
- ⋆ General formula $\alpha=3-2\ln|\det M|/\ln\lambda_1$ for 2×2 Pisot matrices
- ⋆ First $\alpha=2.071$ chain ($\bar{\Lambda}\approx0.650$, rough est.)
- ⋆ Silver $\bar{\Lambda}=1/4$ confirmed at 0.3σ
- ⋆ Metallic-mean limit $= 1/3$ (URL cloaked)

Open questions:

- Analytical proof of $\bar{\Lambda}(\text{Silver}) = 1/4$?
- Precise $\bar{\Lambda}$ for the $\alpha = 2.071$ chain?
- Closed form for $\bar{\Lambda}(\mu_n)$?
- Can we find α closer to 3?

The non-unimodular result is the main new finding: the $\alpha \in (2, 3)$ gap CAN be filled, but only by breaking the unimodular constraint. Silver= $1/4$ would be the first metallic-mean chain with a rational $\bar{\Lambda}$. The limit $1/3 =$ URL cloaked is structurally satisfying: as the chain approaches maximum Class I disorder, it converges to the disordered limit.

References

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